

ENTERPRISE TECHNICAL EVIDENCE

OXYILE FINTECH PLATFORM: TECHNICAL R&D, ARCHITECTURE & EVIDENCE REPORT

Enterprise Technical Whitepaper & Architecture Manual prepared
for Director Review & FCA Regulatory Endorsement Pack (July
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MARKET & CURRENCY

United Kingdom
Strictly British Pounds (£ GBP)

EVIDENCE METHOD

Automated Playwright browser capture
1920 × 1080 Chromium viewport

CLASSIFICATION

Director / Regulatory Pack
Technical R&D evidence

DOCUMENT NAVIGATION

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This whitepaper combines architecture explanation, sequential transaction lifecycles, automated UI evidence and full portal feature catalogues.

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Scope statement. This is a technical R&D and delivery evidence report. It supports—but does not itself constitute—FCA authorisation, endorsement, legal advice, an audit opinion or a production security certification.

CHAPTER 1

Executive Summary & Platform Architecture

Oxyile is a UK-market lending platform combining verified user journeys, bilateral loan handshakes, guarantor-backed payment controls, governed content operations and enterprise people systems in one role-aware Next.js estate.

Next.js Application Layer

App Router pages separate public acquisition, authenticated dashboards, HR, Blogger and Admin workspaces. React server components protect initial data access; server actions and route handlers own mutations, payment calls and service-role operations.

Supabase Data & Identity

Supabase Auth establishes sessions. PostgreSQL stores profiles, handshakes, ATS, HRMS and editorial state. RLS applies row-level boundaries, while server-side assertions enforce ADMIN, HR, BLOGGER, INVESTOR and BORROWER roles.

GoCardless Payment Rail

GBP Billing Requests and Direct Debit mandates are created with server-held credentials. Hosted authorisation, webhook reconciliation, client-money controls and guarantor fallback logic are isolated from browser code.

Polygon Audit Anchor

Canonical handshake facts are hashed and relayed to Polygon Amoy. Confirmed transaction hashes return to Postgres, producing a tamper-evident cross-reference while operational and legal records remain off-chain.

Delivery architecture

The platform follows a defence-in-depth model: middleware gates protected path families; page layouts verify user and profile roles; server actions repeat authorisation; RLS constrains direct browser queries; service-role clients appear only in trusted server modules. UI state is never treated as authority. Every GBP amount is revalidated at a server boundary and payment-provider amounts are converted to integer pence.

Operationally, Supabase Realtime provides low-latency state propagation for handshakes and guarantor status. Polling fallback handles websocket interruption. Webhooks reconcile external payment events into internal state and invoke on-chain audit updates through a controlled relay. The HRMS and Editorial subsystems reuse the same auth, storage and audit patterns without mixing their regulated data domains.

CHAPTER 2

Open Banking Payment Flow, Collateral Enforcement & On-Chain Audit Trail

The programme combines borrower origination, guarantor consent, GoCardless Direct Debit, client-money reconciliation, realtime state and a Polygon audit anchor. The six-step lifecycle below distinguishes implemented controls from regulatory assumptions.

STEP 1

Interactive Loan Proposal & Origination

A verified borrower originates a collateral-backed proposal from the authenticated handshake workspace. The client validates principal and duration; the server receives the proposal and supporting collateral as multipart FormData where files are present. Principal, fixed annual rate and duration are normalised before the EMI calculator derives the monthly amount and total return in British Pounds. The platform supports £120+ ticket requests subject to product and affordability rules. Identity columns bind the borrower and lender to the record, while Supabase RLS limits direct client access to authorised counterparties. Server-only actions use privileged access only after explicit role and ownership checks.

- GBP is stored as numeric values and displayed with en-GB formatting.
- Proposal state starts pending and cannot become active before bilateral approval.
- Collateral type, valuation, description and evidence are required in the borrower path.
- Audit-relevant timestamps are generated server-side.

STEP 2

Co-Applicant / Guarantor Liability Link

The proposal records a guarantor email and pending guarantor status against the handshake. A signed invitation token binds handshake ID, recipient email and issue time, preventing an arbitrary third party from opening the liability page. The guarantor reviews principal, rate, tenure and estimated EMI before acceptance. Consent changes the Postgres handshake state and, once authorised, stores the GoCardless mandate reference. This backup rail reduces expected loss by allowing an agreed EMI collection attempt after a borrower payment failure; it does not transfer ownership of the underlying loan.

- Invite-token verification precedes all mandate setup calls.
- Recipient email must equal the guarantor email persisted on the handshake.
- Acceptance and rejection are explicit states with timestamped database events.
- The liability statement explains that the rail is used only after a missed EMI.

STEP 3**GoCardless Sandbox Direct Debit Mandate Creation**

The server initialises the official GoCardless Node client against Sandbox unless the live environment is explicitly selected. The browser posts JSON to the application; a compatibility parser also accepts legacy multipart form submissions. The application creates a GBP billing request, attaches handshake metadata, creates a billing-request flow and returns the authorisation URL as JSON. The browser then performs a GET navigation—never a 307/POST relay—to the hosted GoCardless authorisation experience. Sandbox values such as sort code 20-00-00 and account number 55779911 are test data only and must never appear in live customer instructions.

Accuracy note. Architecture accuracy note: this implementation is a GoCardless Direct Debit / Billing Request integration, not a generic Open Banking payment initiation API. The chapter title retains the programme terminology while distinguishing the actual payment rail.

- Currency is hard-locked to GBP in the mandate request.
- Access tokens remain server-side and never enter client bundles.
- The callback includes handshake and signed guarantor context.
- Production should validate the GoCardless webhook signature before processing events.

STEP 4**Real-Time Mandate Confirmation via Supabase Realtime**

On completion, the callback reconciles the billing request and writes mandate identifiers plus the accepted guarantor state to Postgres. Chat and handshake components subscribe to the relevant Supabase Realtime channel and also use a bounded polling fallback. A committed row change is broadcast to authorised clients, which re-render the “Guarantor Secured” state without a page reload. The fallback protects UX if a websocket is interrupted, while the database remains the source of truth.

- Realtime is an observation channel; it does not itself authorise a mandate.
- Subscription filters constrain updates to the relevant handshake.
- Polling is time-bounded and retrieves authoritative server state.
- Status transitions are idempotent to tolerate duplicate callbacks.

STEP 5**Investor Escrow Funding & Account Reconciliation**

Once both parties approve, the authorised funding screen retrieves a server-generated handshake preview and verifies that the current user is a valid party. The investor/borrower funding APIs reject missing sessions with HTTP 401 and return structured JSON so the UI can offer a sign-in recovery path instead of silently failing. GoCardless-originated funds are treated as client money: payment references are reconciled to the handshake, AML/sanctions status and settled balance are checked, and funds remain locked until treasury conditions are satisfied.

- Session and ownership checks occur again at every mutating endpoint.
- Amounts convert from pounds to integer pence at the payment boundary.
- Client-money release is separated from payment notification.
- Requests and callbacks use stable handshake/payment references for reconciliation.

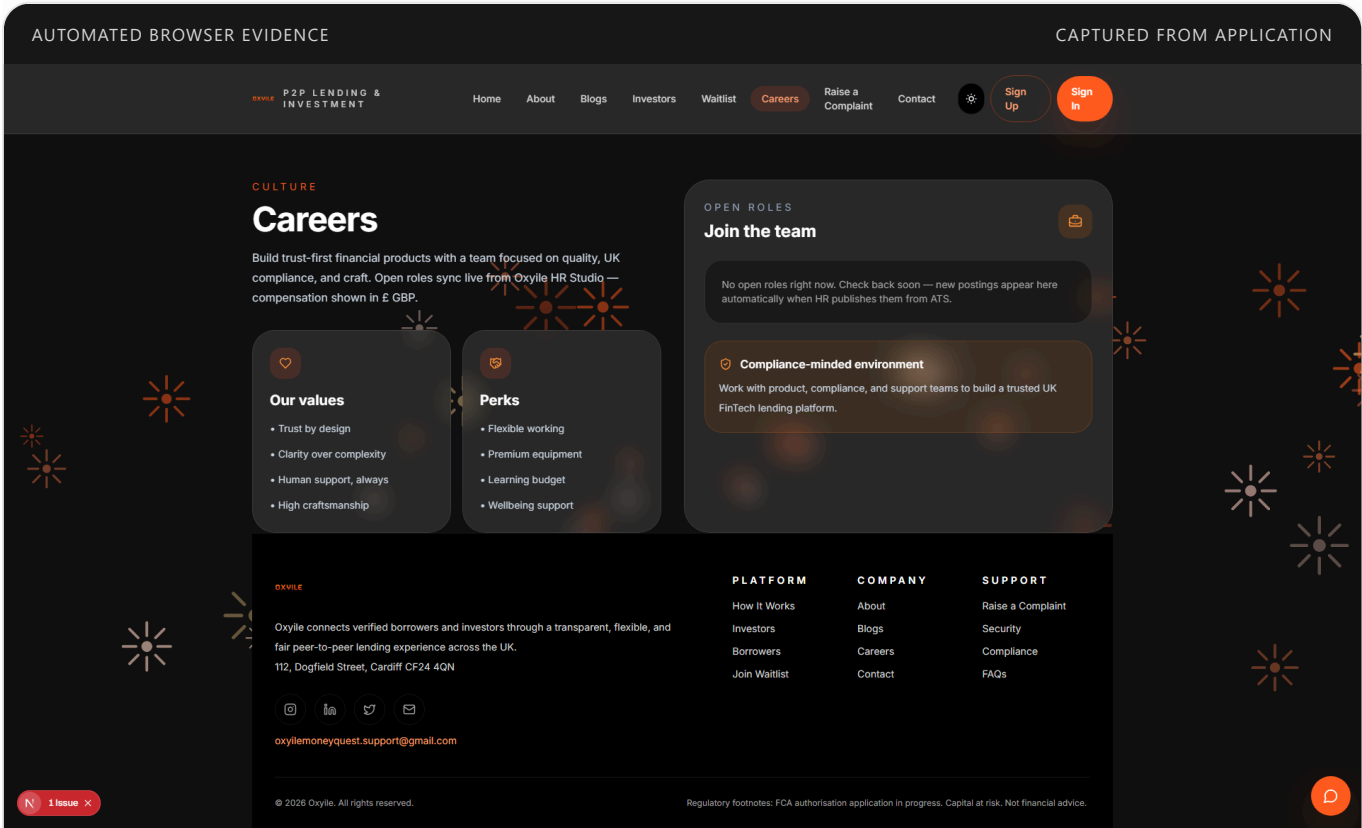
STEP 6**Polygon Blockchain On-Chain Audit Trigger**

After authenticated escrow finalisation, the server hashes canonical handshake facts—IDs, parties, amount, approval timestamp and optional guarantor identity. A server-held relayer wallet submits a zero-value transaction carrying encoded audit data to Polygon Amoy. The action waits for a receipt, validates the 32-byte transaction hash, and only then writes tx_hash / polygon_tx_hash plus activation data to Supabase. Payment webhooks also invoke a queued on-chain EMI update path. The immutable anchor provides tamper-evident evidence; it does not replace the regulated ledger, customer contract or Postgres operational record.

- The Polygon private key is server-only and normalised before wallet construction.
 - Both lender and borrower approvals are prerequisites.
 - Database sync occurs after chain confirmation; partial-success errors preserve the hash.
 - Unavailable chain writes can be queued for controlled retry.
-

CHAPTER 3

Automated Product Evidence



Dynamic UK Careers Portal. Live public job feed synchronised from ATS postings, showing employment metadata and £ GBP salary bands. Where an open role exists, the script opens its detailed application drawer.

AUTOMATED BROWSER EVIDENCE

AUTHENTICATION EVIDENCE PENDING

PROTECTED EVIDENCE CAPTURE

Editorial Studio & SEO Guide

Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

<http://127.0.0.1:3000/blogger/seo-guide>

Editorial Studio & SEO Guide. Governed content workspace documenting the SEO engine, rich editing, admin feedback and distribution controls.

Capture note: Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

CHAPTER 3 · CONTINUED

Enterprise People & Recruitment Evidence

AUTOMATED BROWSER EVIDENCE

AUTHENTICATION EVIDENCE PENDING

PROTECTED EVIDENCE CAPTURE

ATS Kanban Hiring Board

Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

`http://127.0.0.1:3000/hr/recruitment`

ATS Kanban Hiring Board. Enterprise recruitment pipeline with AI match scores, DBS status, stage governance, source analytics and £ GBP offer workflow.

Capture note: Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

AUTOMATED BROWSER EVIDENCE

AUTHENTICATION EVIDENCE PENDING

PROTECTED EVIDENCE CAPTURE

HRMS Compliance & People Operations

Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

http://127.0.0.1:3000/hr/employees

HRMS Compliance & People Operations. Employee compliance, leave, FCA training, NDA, probation, burnout, assets and access-provisioning controls.

Capture note: Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

CHAPTER 4

Oxyile Editorial Studio & SEO Content Engine

The Editorial Studio combines semantic writing, workflow governance, real-time on-page analysis and multi-channel reuse. Feature records below are generated from the same catalogue that powers the in-product SEO Guide, supplemented by the editorial controls implemented around it.

System flow. Draft → semantic rich-text authoring → live SEO checks → inline media upload → admin review/edit → approve or structured reject → blogger remediation → resubmit → publish and repurpose.

SEO-1

EDITORIAL WORKFLOW

Notion-Style TipTap Rich-Text Editor

Purpose and implementation. A structured document editor built on TipTap extensions rather than a plain textarea. It supports headings, lists, emphasis, hyperlinks and image nodes while retaining semantic HTML for crawlability.

Operator sequence. Authors compose in blocks, type "/" to open slash commands, and use the toolbar only when explicit formatting is needed.

Control location. Blogger → Blog Editor and the admin review editor

Business / regulatory value. Consistent semantic markup improves accessibility, editorial speed and search-engine interpretation.

SEO-2

EDITORIAL WORKFLOW

Slash Command Palette

Purpose and implementation. A keyboard-first insertion menu for headings, lists, quotes and image blocks, modelled on modern collaborative writing tools.

Operator sequence. Type "/", filter a command and select it without leaving the writing flow.

Control location. Inside the rich-text body editor

Business / regulatory value. Reduces formatting friction and increases document consistency.

SEO-3

EDITORIAL WORKFLOW

Inline Multi-Image Storage Pipeline

Purpose and implementation. Multiple body images upload to a dedicated Supabase Storage bucket and are represented as TipTap image nodes plus a JSONB URL index.

Operator sequence. Choose a JPG, PNG, WebP or GIF; the server action uploads it and inserts the public URL at the current cursor.

Control location. Editor "+ image" action between paragraphs

Business / regulatory value. Creates visually rich long-form content without conflating inline media with the cover asset.

SEO-4

GOVERNANCE

Admin Rejection & Structured Feedback Loop

Purpose and implementation. Reviewers select a rejection category and attach exact changes requested. The reason and feedback persist with the post.

Operator sequence. Select Plagiarism, Formatting, Poor SEO, Tone, Accuracy or Other; write actionable changes; submit rejection.

Control location. Admin Blog Manager → pending review → Reject

Business / regulatory value. Creates a traceable editorial control and reduces ambiguous revision cycles.

SEO-5

GOVERNANCE

Admin Edit-Before-Approval

Purpose and implementation. Administrators use the same rich editor as bloggers to correct content directly before approval.

Operator sequence. Open a pending article, edit body and images, save, then approve.

Control location. Admin Blog Manager review panel

Business / regulatory value. Removes copy-and-paste workflows and preserves a single canonical record.

SEO-6

GOVERNANCE

Rejected-Post Resubmission

Purpose and implementation. Rejected cards display a red banner and the full admin feedback appears above the editor.

Operator sequence. Open “Fix & Resubmit”, address each request and return the post to approval.

Control location. Blogger → My Drafts / Rejected

Business / regulatory value. Makes remediation status visible and auditable.

SEO-7

EXPERIENCE

Zero-Flicker Loading & Tab Isolation

Purpose and implementation. Tab transitions clear stale arrays immediately, cancel superseded fetches and render layout-matched pulse skeletons before empty states.

Operator sequence. Automatic whenever the active tab changes or data refreshes.

Control location. Drafts, Pending, Published and References tabs

Business / regulatory value. Prevents previous-tab state leakage and misleading “No blogs” flashes.

SEO-8

EXPERIENCE

Persistent Editorial Quick Create

Purpose and implementation. A glassmorphic bottom navigation provides one-click access to a standard blog or an SEO-scored draft.

Operator sequence. Press the central orange Create action and choose the desired workflow.

Control location. Every Blogger Portal route

Business / regulatory value. Shortens the path from idea to governed content.

**RESEARCH****Keyword Research Hub**

Purpose and implementation. Search volume, competition, long-tail and LSI seeds for FinTech topics.

Operator sequence. Enter a keyword → Run research → Draft from keyword.

Control location. Blogger → SEO Studio → Keyword Hub tab

Business / regulatory value. Targets demand with realistic competition before you write.

**RESEARCH****Topic Suggestion Engine**

Purpose and implementation. Trending UK lending / FinTech angles ready to draft.

Operator sequence. Click a topic card to open a pre-titled SEO draft.

Control location. SEO Studio → Topic Engine

Business / regulatory value. Cuts ideation time and stays niche-relevant.

**RESEARCH****Competitor Analysis**

Purpose and implementation. Competitor URLs + content gaps for a focus keyword.

Operator sequence. Enter keyword → Analyse gaps → Cover missing angles in your outline.

Control location. SEO Studio → Competitors

Business / regulatory value. Wins rankings by filling what competitors skip.

**ON-PAGE****Real-Time On-Page SEO Checklist**

Purpose and implementation. Live checks for title, meta, H2s, density, links, length.

Operator sequence. Write while watching green/red checks; fix failing items before submit.

Control location. SEO draft studio right sidebar

Business / regulatory value. Prevents publishing thin or under-optimized pages.

**ON-PAGE****Readability Scorer**

Purpose and implementation. Flags complex sentences, passive voice, and jargon.

Operator sequence. Rewrite flagged sentences into shorter active voice.

Control location. SEO studio → Readability scorer panel

Business / regulatory value. Higher engagement and better voice-search answers.

**ON-PAGE****SERP Preview**

Purpose and implementation. Desktop/mobile Google result mock for title + meta.

Operator sequence. Toggle Desktop/Mobile and adjust title/meta until it looks clickable.

Control location. SEO studio main column

Business / regulatory value. Improves CTR before you ever rank.

links

ON-PAGE**Link Suggestion Engine**

Purpose and implementation. Internal pages + high-authority external domains.

Operator sequence. Click suggestions to insert anchors into the body.

Control location. SEO studio → Link suggestions

Business / regulatory value. Builds topical authority and trust signals.

copilot

AI ASSIST**AI Content Autocomplete**

Purpose and implementation. Shift+Tab continues your thought in FinTech voice.

Operator sequence. Write a lead-in, press Shift+Tab, edit the inserted paragraph.

Control location. SEO studio editor

Business / regulatory value. Breaks writer's block without leaving the page.

lsi

AI ASSIST**LSI Keyword Injector**

Purpose and implementation. Semantic terms to sprinkle for topical depth.

Operator sequence. Click a chip to insert a natural LSI prompt paragraph.

Control location. SEO studio → LSI injector

Business / regulatory value. Covers related entities Google expects.

trust

AI ASSIST**Plagiarism & AI Trust Score**

Purpose and implementation. Human-like trust heuristic from jargon/links/depth.

Operator sequence. Raise trust by adding sources and cutting buzzwords.

Control location. Live SEO scores → Trust / human

Business / regulatory value. Signals original, compliant editorial quality.

social

DISTRIBUTION**Social Media Repurposing**

Purpose and implementation. One-click Twitter thread + LinkedIn post.

Operator sequence. Generate, copy, schedule on social.

Control location. SEO studio → Social repurposing

Business / regulatory value. Amplifies distribution without rewriting.

voice

ADVANCED**Voice Search Analyzer**

Purpose and implementation. Scores how well content answers spoken queries.

Operator sequence. Add concise Q&A style H2s and short answer paragraphs.

Control location. Live scores → Voice search

Business / regulatory value. Captures Siri/Alexa-style long-tail traffic.

**ADVANCED****Featured Snippet Optimizer**

Purpose and implementation. Flags lists/definitions for Position Zero.

Operator sequence. Add a 40-word definition or numbered steps near the top.

Control location. Snippet + voice panel

Business / regulatory value. Competes for Google's featured answer box.

**ADVANCED****Content Decay Alerts**

Purpose and implementation. Surfaces drafts untouched for 90+ days.

Operator sequence. Open aging posts, refresh stats/examples, republish.

Control location. SEO Studio → Decay Alerts

Business / regulatory value. Protects rankings before traffic slips.

**ADVANCED****A/B Headline Tester**

Purpose and implementation. Scores 3 title variants for predicted CTR.

Operator sequence. Enter variants → Use the top scorer as your title.

Control location. SEO studio → A/B headline tester

Business / regulatory value. Data-informed headlines beat guesswork.

**ADVANCED****Read-Time & Engagement Predictor**

Purpose and implementation. Estimated minutes and scroll-depth prediction.

Operator sequence. Aim for clear H2 structure to lift predicted depth.

Control location. Editor status bar + live scores

Business / regulatory value. Aligns length with audience attention.

**ADVANCED****Cannibalization Warning**

Purpose and implementation. Alerts if another draft shares the same focus keyword.

Operator sequence. Differentiate keywords or merge overlapping drafts.

Control location. Yellow banner under focus keyword

Business / regulatory value. Stops your own pages competing with each other.

**ON-PAGE****Sticky Table of Contents**

Purpose and implementation. Auto H2/H3 outline.

Operator sequence. Add headings; TOC updates live.

Control location. SEO studio → TOC card

Business / regulatory value. Improves UX and crawlable structure.

alt

ON-PAGE**Image Alt-Text AI****Purpose and implementation.** Generates SEO-friendly alt text.**Operator sequence.** Click Generate, then tweak for accuracy.**Control location.** SEO studio → Image alt AI**Business / regulatory value.** Accessibility + image search visibility.

i18n

ADVANCED**Multilingual SEO Expanding****Purpose and implementation.** Regions where the topic is underserved.**Operator sequence.** Prioritize locales for future translations.**Control location.** Multilingual SEO panel**Business / regulatory value.** Opens international long-tail demand.

graph

ADVANCED**Dynamic Internal Link Graph****Purpose and implementation.** Visual map of links from this draft.**Operator sequence.** Add internal links until the graph fills out.**Control location.** Internal link graph panel**Business / regulatory value.** Strengthens site architecture.

evergre

ADVANCED**Evergreen vs Trending Tag****Purpose and implementation.** Classifies pillar vs news-style content.**Operator sequence.** Adjust wording if you want evergreen longevity.**Control location.** Top badge on SEO studio**Business / regulatory value.** Sets the right update cadence.

tone

NEW 2026 TOOLS**Tone & Sentiment Analyzer****Purpose and implementation.** Detects Authoritative, Conversational, or Urgent tone.**Operator sequence.** Rewrite until tone matches the audience intent.**Control location.** SEO studio → New tools panel**Business / regulatory value.** Matches SERP intent and brand voice.

newslet

NEW 2026 TOOLS**Newsletter One-Click Convert****Purpose and implementation.** Turns a long post into a ~300-word email.**Operator sequence.** Generate, copy into Admin Newsletter broadcast.**Control location.** New tools → Newsletter convert**Business / regulatory value.** Recycles SEO content into owned channels.

flesch

NEW 2026 TOOLS

Reading Grade Level Scorer**Purpose and implementation.** Flesch-Kincaid grade aiming for ~8th grade.**Operator sequence.** Shorten sentences until grade ≤ 8 .**Control location.** New tools → Grade level**Business / regulatory value.** Broader comprehension = more engagement.

tweets

NEW 2026 TOOLS

Auto-Tweet Quote Extractor**Purpose and implementation.** Highlights 3 tweetable quotes with copy.**Operator sequence.** Copy the best line into X/LinkedIn.**Control location.** New tools → Tweet quotes**Business / regulatory value.** Faster social syndication.meta-
nb

NEW 2026 TOOLS

Meta Tags A/B Generator**Purpose and implementation.** 3 SEO titles + 3 meta descriptions.**Operator sequence.** Apply a variant to title/meta fields.**Control location.** New tools → Meta A/B**Business / regulatory value.** Higher SERP CTR experiments.

heatma

NEW 2026 TOOLS

Keyword Density Heatmap**Purpose and implementation.** Warns when focus keyword is overused.**Operator sequence.** Thin repetitions if density $> 2.5\%$.**Control location.** New tools → Density heatmap**Business / regulatory value.** Avoids keyword stuffing penalties.

orphan

NEW 2026 TOOLS

Orphan Page Detector**Purpose and implementation.** Warns if the draft lacks internal link support.**Operator sequence.** Add hub links from/to related posts.**Control location.** New tools → Orphan alert**Business / regulatory value.** Keeps new URLs discoverable.

authori

NEW 2026 TOOLS

External Authority Scraper**Purpose and implementation.** Suggests high-DR sites to cite.**Operator sequence.** Insert 1–2 citations with descriptive anchors.**Control location.** New tools → Authority links**Business / regulatory value.** E-E-A-T and topical trust.



NEW 2026 TOOLS

Grammar & Formatting Magic Wand

Purpose and implementation. Fixes double spaces and title-cases H2/H3.

Operator sequence. One click, then review headings.

Control location. New tools → Magic wand

Business / regulatory value. Cleaner crawlable structure.



NEW 2026 TOOLS

Estimated Traffic Value Calculator

Purpose and implementation. Mock £/mo value from volume × score × CPC (GBP).

Operator sequence. Use to prioritize which draft to finish first.

Control location. New tools → Traffic value

Business / regulatory value. Aligns editorial effort with ROI.

CHAPTER 5

Enterprise HRMS & ATS Portal

The people platform standardises recruitment, compliance, attendance, leave, compensation and executive intelligence around British Pounds and UK operating expectations. The public careers form writes directly into the Applied stage of the ATS.

System flow. Headcount request → budget approval → public job sync → applicant upload → AI match → stage governance → DBS / scorecard → GBP offer → onboarding → attendance, payroll, performance and offboarding evidence.



ATS & RECRUITMENT

Visual Kanban Hiring Board

Purpose and implementation. Drag or move applicants across Applied → Hired/Rejected. Public /careers applications land in Applied automatically.

Operator sequence. Open a card → change stage via column actions or drag handle. Refresh after a careers apply to see the new candidate.

Control location. HR → ATS Recruitment

Business / regulatory value. Transparent pipeline visibility for regulated hiring decisions.

2

ATS & RECRUITMENT**AI Resume Score & Matcher**

Purpose and implementation. Scores resumes 0–100% against job requirements and AI Match Keywords from the Enterprise Job Editor.

Operator sequence. In Enterprise Job Editor, add comma-separated AI Match Keywords → publish. New /careers applicants are scored on insert.

Control location. ATS board — score badge on each card; keywords set when creating a job

Business / regulatory value. Faster shortlists without bias toward formatting alone.

3

ATS & RECRUITMENT**UK Background & DBS Check Tracker**

Purpose and implementation. Tracks financial/criminal verification status.

Operator sequence. Cycle status: not started → DBS pending → clear/flagged.

Control location. Applicant card → Background badge

Business / regulatory value. Supports FCA fitness & propriety expectations.

4

ATS & RECRUITMENT**One-Click Interview Scheduler**

Purpose and implementation. Schedule calls and capture interview notes.

Operator sequence. Pick datetime → save notes → stage moves to Interview.

Control location. ATS → Schedule Interview modal

Business / regulatory value. Audit-ready interview trail.

5

ATS & RECRUITMENT**Automated Email Templates**

Purpose and implementation. Interview invite or polite rejection copy.

Operator sequence. Click Invite or Reject email → copy/send via mailto.

Control location. ATS → Email actions on card

Business / regulatory value. Consistent, professional candidate communications.

6

ATS & RECRUITMENT**Offer Letter Generator**

Purpose and implementation. Digital offer with £ GBP salary & PAYE snapshot.

Operator sequence. Enter salary GBP → generate HTML letter → move to Offer stage.

Control location. ATS → Generate Offer

Business / regulatory value. Clear compensation transparency in UK terms.

7

ATS & RECRUITMENT

Candidate Scorecards**Purpose and implementation.** Structured tech + culture evaluation.**Operator sequence.** Rate technical/culture fit 1–5 and save notes.**Control location.** ATS → Scorecard panel**Business / regulatory value.** Defensible hiring decisions.

8

ATS & RECRUITMENT

Duplicate Applicant Flag**Purpose and implementation.** Alerts if email applied before.**Operator sequence.** System flags on create when email already exists.**Control location.** ATS card — Duplicate badge**Business / regulatory value.** Prevents double-processing and policy gaming.

9

ATS & RECRUITMENT

Talent Pool Archive**Purpose and implementation.** Save strong second-choice candidates.**Operator sequence.** Toggle talent pool; filter Talent Pool tab.**Control location.** ATS → Add to Talent Pool**Business / regulatory value.** Reuse warm talent for future requisitions.

10

ATS & RECRUITMENT

Source of Hire Analytics**Purpose and implementation.** Chart LinkedIn / Direct / Referral sources.**Operator sequence.** Set source on applicants; review distribution chart.**Control location.** ATS → Analytics strip**Business / regulatory value.** Optimise recruiting spend.

11

ONBOARDING & ATTENDANCE

FCA Compliance Onboarding Checklist**Purpose and implementation.** Track mandatory UK compliance training.**Operator sequence.** Mark FCA trained when modules complete.**Control location.** Employees → Compliance column**Business / regulatory value.** Demonstrates staff competence for regulated activities.

12

ONBOARDING & ATTENDANCE

Smart Remote Attendance Log**Purpose and implementation.** Check-in/out with IP/location tags.**Operator sequence.** Log check-in; later check-out with optional location.**Control location.** HR → Attendance**Business / regulatory value.** Evidence for remote-working policies.

13

ONBOARDING & ATTENDANCE**Leave Management Engine**

Purpose and implementation. Sick/Casual/Annual leave with 1-click approve.

Operator sequence. Submit leave → HR Approves or Rejects.

Control location. Employees & Leaves / Quick Create

Business / regulatory value. Fair, auditable time-off governance.

14

ONBOARDING & ATTENDANCE**UK Public Holiday Calendar**

Purpose and implementation. Highlight England & Wales bank holidays.

Operator sequence. Toggle calendar to plan coverage.

Control location. Attendance → UK Holidays toggle

Business / regulatory value. Avoid scheduling conflicts on bank holidays.

15

ONBOARDING & ATTENDANCE**NDA & Policy Acknowledgement Vault**

Purpose and implementation. Track signed confidentiality agreements.

Operator sequence. Mark NDA signed after DocuSign/vault upload.

Control location. Employees → NDA badge

Business / regulatory value. Protects lender IP and customer data.

16

ONBOARDING & ATTENDANCE**Probation Monitoring**

Purpose and implementation. Flags 30/60/90 day reviews.

Operator sequence. Set probation dates; review flagged staff weekly.

Control location. Employees → Probation alerts

Business / regulatory value. Structured performance during probation.

17

ONBOARDING & ATTENDANCE**Overtime Tracker**

Purpose and implementation. Extra hours with manager sign-off.

Operator sequence. Log hours → manager signs off for payroll.

Control location. Attendance → Overtime

Business / regulatory value. Accurate GBP payroll adjustments.

18

ONBOARDING & ATTENDANCE**Burnout Risk Indicator**

Purpose and implementation. Flags no leave 90+ days or chronic OT.

Operator sequence. Review flagged people; encourage leave.

Control location. Employees → Burnout badge

Business / regulatory value. Duty of care and retention.

19

ONBOARDING & ATTENDANCE**Asset Allocation Tracker****Purpose and implementation.** Laptops, keys, test devices.**Operator sequence.** Allocate asset → collect on offboarding.**Control location.** Employees → Assets**Business / regulatory value.** Security & inventory control.

20

ONBOARDING & ATTENDANCE**Role Access Provisioning Request****Purpose and implementation.** Ticket tech to grant/revoke admin access.**Operator sequence.** Submit grant/revoke → Admin Access team completes.**Control location.** Employees → Access request**Business / regulatory value.** Least-privilege access discipline.

21

PAYROLL & PERFORMANCE**UK Salary & Payslip Vault****Purpose and implementation.** Basic, HRA, pension, NI in £ GBP.**Operator sequence.** Open employee → review breakdown.**Control location.** Payroll £**Business / regulatory value.** Transparent UK compensation records.

22

PAYROLL & PERFORMANCE**Estimated PAYE Viewer****Purpose and implementation.** Preview UK tax bands & NI.**Operator sequence.** Select employee gross → view tax/NI/net.**Control location.** Payroll → PAYE estimate**Business / regulatory value.** Helps explain take-home to staff.

23

PAYROLL & PERFORMANCE**Expense Reimbursement Portal****Purpose and implementation.** Travel/software claims with receipts.**Operator sequence.** Submit claim → HR approve (exec if >£500).**Control location.** Payroll / Quick Create → Expense**Business / regulatory value.** Controlled spend with audit trail.

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PAYROLL & PERFORMANCE**Quarterly KPI & OKR Tracker****Purpose and implementation.** Goals with progress bars.**Operator sequence.** Add quarterly goal → update % progress.**Control location.** Performance**Business / regulatory value.** Aligns teams to lender OKRs.

25

PAYROLL & PERFORMANCE

360° Peer Feedback**Purpose and implementation.** Manager/peer appraisal notes.**Operator sequence.** Log rating + constructive note.**Control location.** Performance → Feedback**Business / regulatory value.** Richer performance evidence.

26

PAYROLL & PERFORMANCE

Incentive & Bonus Calculator**Purpose and implementation.** £ bonus from KPI achievement.**Operator sequence.** KPI score drives suggested bonus GBP.**Control location.** Performance → Bonus**Business / regulatory value.** Fair, formula-based incentives.

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PAYROLL & PERFORMANCE

Contractor vs FTE Tagging**Purpose and implementation.** Classify for tax/regulatory treatment.**Operator sequence.** Tag full_time or contractor on profile.**Control location.** Employees → Employment type**Business / regulatory value.** IR35 / employment status clarity.

28

PAYROLL & PERFORMANCE

Anonymous Grievance Box**Purpose and implementation.** Secure whistleblower reporting.**Operator sequence.** Submit anonymous subject/body.**Control location.** Performance → Grievance**Business / regulatory value.** SMCR / speak-up culture compliance.

29

PAYROLL & PERFORMANCE

Offboarding & Exit Interview**Purpose and implementation.** Revoke access, collect assets, notes.**Operator sequence.** Start checklist → complete exit interview.**Control location.** Employees → Offboard**Business / regulatory value.** Secure leaver process.

30

PAYROLL & PERFORMANCE

Audit Trail Export**Purpose and implementation.** Export HR actions to CSV.**Operator sequence.** Click Export CSV for regulators.**Control location.** Payroll / Guide → Export Audit**Business / regulatory value.** Ready evidence for audits.

31

EXECUTIVE INTELLIGENCE

Monthly Headcount Budget Burn**Purpose and implementation.** Total monthly payroll £ spend.**Operator sequence.** Open widget for burn rate.**Control location.** Admin → HR Overview**Business / regulatory value.** Executive cost control.

32

EXECUTIVE INTELLIGENCE

Attrition & Retention Risk**Purpose and implementation.** AI-style department turnover risk.**Operator sequence.** Review risk score; engage managers.**Control location.** Admin HR Overview**Business / regulatory value.** Protect critical FinTech talent.

33

EXECUTIVE INTELLIGENCE

Budget Approval Workflow**Purpose and implementation.** Admin approves headcount budget.**Operator sequence.** HR requests → Admin Approves.**Control location.** Admin HR Overview → Headcount**Business / regulatory value.** Prevents unbudgeted hiring.

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EXECUTIVE INTELLIGENCE

Executive Headcount Breakdown**Purpose and implementation.** Employees, contractors, vacancies.**Operator sequence.** Glance live counters.**Control location.** Admin HR Overview counters**Business / regulatory value.** Instant workforce composition.

35

EXECUTIVE INTELLIGENCE

Department Cost Center Chart**Purpose and implementation.** Tech vs Marketing vs Ops salary spend.**Operator sequence.** Compare department GBP spend.**Control location.** Admin HR Overview chart**Business / regulatory value.** Allocate budget to growth areas.

36

EXECUTIVE INTELLIGENCE

High-Performer Leaderboard**Purpose and implementation.** Top KPI achievers for execs.**Operator sequence.** Celebrate / retain top talent.**Control location.** Admin HR Overview**Business / regulatory value.** Culture of excellence.

37

EXECUTIVE INTELLIGENCE

Pending Critical Actions

Purpose and implementation. Expense >£500 or salary increments awaiting sign-off.

Operator sequence. Clear critical queue weekly.

Control location. Admin HR Overview alerts

Business / regulatory value. No stalled high-value approvals.

38

EXECUTIVE INTELLIGENCE

Internal Referral Bonus Tracker

Purpose and implementation. £ referral payouts when hires succeed.

Operator sequence. Track pending → mark paid.

Control location. Admin / ATS referrals

Business / regulatory value. Incentivises quality referrals.

39

EXECUTIVE INTELLIGENCE

Anniversary & Birthday Widget

Purpose and implementation. Upcoming culture milestones.

Operator sequence. Send recognition notes.

Control location. Admin HR Overview

Business / regulatory value. Builds belonging.

40

EXECUTIVE INTELLIGENCE

Zero-Flicker Skeleton Loading

Purpose and implementation. animate-pulse skeletons while fetching.

Operator sequence. Automatic during loads.

Control location. All HR tabs

Business / regulatory value. Premium, trustworthy UX.

41

ATS & RECRUITMENT

Public Careers Sync (/careers)

Purpose and implementation. Open ATS job postings with publish enabled automatically appear on the public /careers page as glassmorphic role cards (£ GBP salary bands).

Operator sequence. Create a job in the Enterprise Editor, set £ min-max, tick "Publish to ATS as Open & sync to /careers", ensure HR Settings sync toggle is on. Visit /careers to confirm the listing.

Control location. HR Settings → Public careers sync · ATS → Enterprise Job Editor → Publish checkbox

Business / regulatory value. Single source of truth — no duplicate marketing copy for open roles.

42

ATS & RECRUITMENT

Enterprise Job Editor

Purpose and implementation. Spacious 2-column creator for title, department, employment type, location, £ GBP salary band, description, FCA responsibilities, and AI Match Keywords.

Operator sequence. Fill metadata on the left, rich details on the right, add comma-separated AI keywords, then create. Use polished validation messages if fields are missing.

Control location. ATS Recruitment → + Create job (or Quick Create → Post New Job)

Business / regulatory value. Detail-rich requisitions improve matcher quality and candidate clarity.

43

ATS & RECRUITMENT

Careers → ATS Application Pipeline

Purpose and implementation. Candidates open a job detail drawer on /careers, apply with name, email, LinkedIn, and PDF resume; the record is inserted into job_applicants at Applied with an AI match score.

Operator sequence. After a public apply, open ATS Recruitment — the candidate appears in Applied linked to the job, ready for shortlist / interview.

Control location. Public /careers → View role → Apply Now · HR → ATS Kanban (Applied column)

Business / regulatory value. Zero manual re-entry from careers inbox into the hiring board.

CHAPTER 6

Lending, Guarantor & Funding Evidence

AUTOMATED BROWSER EVIDENCE

AUTHENTICATION EVIDENCE PENDING

PROTECTED EVIDENCE CAPTURE

Investor / Borrower Lending Dashboard

Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

<http://127.0.0.1:3000/dashboard/marketplace>

Investor / Borrower Lending Dashboard. Authenticated lending workspace used to discover counterparties, originate a handshake and progress a collateral-backed GBP facility.

Capture note: Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

AUTOMATED BROWSER EVIDENCE

AUTHENTICATION EVIDENCE PENDING

PROTECTED EVIDENCE CAPTURE

GoCardless & Escrow Funding Flow

Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

<http://127.0.0.1:3000/payments/sandbox>

GoCardless & Escrow Funding Flow. Direct Debit / client-money funding screen. For a live evidence capture, provide REPORT_GOCARDLESS_URL or REPORT_HANDSHAKE_ID for an authorised test handshake.

Capture note: Protected route: provide REPORT_TEST_EMAIL and REPORT_TEST_PASSWORD, REPORT_AUTH_COOKIES_JSON, or REPORT_AUTH_STORAGE_STATE.

Evidence caveat. Guarantor and escrow pages require a valid signed invite or authorised handshake. The automation accepts REPORT_GOCARDLESS_URL / REPORT_HANDSHAKE_ID so the evidence pack can be regenerated against a controlled sandbox record without embedding customer identifiers in source control.

CHAPTER 7

FinTech Security, Database Governance & UK Privacy

Control domain	Technical implementation and assurance purpose
Authentication	Supabase Auth issues secure sessions; protected pages retrieve the authenticated user server-side. Passwords are processed by the identity provider, not stored in application tables.
RBAC	Profile roles plus controlled staff-email/directory mappings route users into Admin, HR, Blogger, Investor or Borrower workspaces. Server assertions repeat checks before privileged operations.
Row-Level Security	RLS policies constrain direct authenticated and anonymous access. Ownership and participant IDs scope borrower, investor, editorial and people records. Service-role use is confined to trusted server code.
Secrets	Supabase service role, GoCardless access token, webhook secret and Polygon private key remain environment variables. The report script accepts auth only through runtime environment or an ignored storage-state file.
Payment integrity	GBP values convert to integer pence at provider boundaries. Hosted mandate authorisation limits account-data handling. Webhooks correlate stable provider IDs with internal handshakes.
Auditability	Postgres timestamps and HR audit logs record mutable operations. Polygon hashes create tamper-evident anchors; transaction hashes cross-reference internal records.
Storage	Dedicated Supabase buckets separate resumes, collateral, covers and inline editorial images. File type and size validation occurs before upload.
UK data privacy	Purpose limitation, access restriction, retention schedules, subject-right handling and deletion procedures remain organisational obligations. RLS and least privilege support those controls but do not replace a DPIA or retention policy.
Availability & UX	Realtime subscriptions use polling fallback. Loading skeletons prevent stale state leakage; structured API errors preserve recoverability after session expiry or provider failure.
Evidence reproducibility	This script captures deterministic 1920×1080 views, embeds image bytes into a self-contained HTML artefact, prints via Chromium and counts final PDF pages with pdf-lib.

Recommended pre-production assurance

Before live regulated use, complete independent penetration testing, webhook signature verification review, RLS policy testing by role, DPIA and retention schedules, incident-response exercises, key-rotation procedures, GoCardless live-environment verification, client-money legal review, vulnerability remediation and an FCA counsel review of customer communications.

CHAPTER 8

R&D Conclusions & Evidence Statement

The delivered platform demonstrates integration R&D across authenticated marketplace interactions, collateral-backed loan negotiation, guarantor liability, external Direct Debit authorisation, realtime data propagation, on-chain anchoring, governed SEO production and enterprise HR operations. The principal engineering achievement is not any single screen; it is the consistent control plane connecting identity, role, state transition, external-provider reconciliation and human-readable evidence.

The automation itself turns that delivery into repeatable evidence. It can run locally or in CI, consume a controlled test identity, capture current interfaces, regenerate an executive whitepaper and report the exact page count and output path. Protected screens are never bypassed: when authentication or a signed handshake is absent, the output explicitly records the missing prerequisite rather than fabricating evidence.

Formal limitation. Statements in this report describe code paths and intended controls observed in the July 2026 R&D build. They are not representations that Oxyile is FCA-authorised or endorsed, that a control has operated effectively over a period, or that Polygon anchoring establishes legal finality. Directors should pair this pack with policy documents, test results, provider agreements, DPIA, financial controls and external legal/compliance review.